When Operator Accuracy Does Not Transfer to the Image: A Case Study of a Learned Interior Transport Operator for Faceted Gemstones

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***Abstract*—A diamond's appearance is made inside it: light refracts in, reflects between facets and leaves in sharply defined directions. Neural appearance models compress such internal light transport into a small network and work well for cloth, where the structure is far smaller than a pixel. This study asks whether the idea transfers to faceted gemstones, where the structure *is* the appearance. A radiance distribution map (RDM) adapted from the cloth method is first shown to condition on the wrong variable, and is generalised to a *learned interior operator*: a network that takes where, in which direction and at what wavelength light enters the stone, and predicts whether, through which facet and in which direction it leaves. Evaluated on three diamond cuts against an exact spectral path tracer, the operator predicts exit facets better than every lookup baseline, yet a more accurate operator produces images that agree with the reference *less* (pixel correlation 0.038–0.168 against 0.136–0.293). A recorded, convergence-trained run on an independent test pool locates the failure: exit facets are predicted correctly for 61% of short paths but at most 7% of paths meeting four or more facets, and for 6% of exits through the pavilion. Each learned decision is 3.5 times faster than tracing, but the neural image settles at an error a single analytic render already beats.**

***Index Terms*—neural rendering, light transport, gemstones, mixture density networks, image evaluation, Monte Carlo rendering**

I. Introduction

The appearance of a faceted diamond depends almost entirely on where light leaves it. Light that enters the crown is refracted, reflected between facets several times and refracted out again, and each facet acts as a mirror onto a different part of the surroundings. Small changes in where light enters move these reflections across facet edges, which is what produces brilliance, fire and sparkle. Path tracing reproduces this exactly but repeats the whole interior journey for every sample.

Neural appearance models offer a way to learn such journeys once and reuse them. Soh and Montazeri [1] compress the light scattered inside cloth fibres into small networks trained from a radiance distribution map (RDM): a table, gathered by tracing the true transport, of how light leaves for each incoming direction. For cloth this is well founded: hundreds of fibres sit inside one pixel, so light enters and leaves within a sub-pixel cross-section and direction alone describes it. A diamond inverts this. It has a few dozen facets, each covering thousands of pixels, and *where* light enters decides which facet it leaves through.

This study asks: *can a compact learned operator reproduce the interior light transport of a faceted gemstone, and if not, which of its errors explain the difference in the image?* Three hypotheses make the question testable:

- **H1 (dissociation).** An operator can improve its exit-facet accuracy without improving the image's agreement with the exact reference.
- **H2 (localisation).** Its errors concentrate in long internal routes and pavilion exits, not in facets that lack training data.
- **H3 (cost).** A faster interior decision does not make the image cheaper at equal quality.

All three are tested under one studio lighting rig on three procedural cuts; the conclusions concern this setting rather than all learned representations of specular media.

II. Background

A. Light transport in a diamond

When light meets a diamond's surface, part of it reflects and the rest refracts into the stone. The split is given by the Fresnel equations and the bend by Snell's law, $n_1 \sin\theta_1 = n_2 \sin\theta_2$, with refractive index $n \approx 2.42$ for diamond. Inside, light meeting a facet beyond the critical angle of about $24.4°$ is totally internally reflected, so it may bounce many times before escaping. The index also varies with wavelength (dispersion), which splits white light into the colours known as fire. A cut is designed so that light entering the crown is reflected by the pavilion back out through the crown; the result is a pattern of sharp reflections whose positions depend sensitively on the viewpoint and on where light enters.

B. Path tracing and Monte Carlo noise

Physically based renderers estimate the light reaching each pixel by tracing random light paths [2], [3]. Each pixel averages many samples per pixel (spp); the estimate is unbiased but noisy, with variance that falls in proportion to $1/N$ for N samples. Two renders with different random seeds therefore differ even when the method is exact, and any comparison between images must account for this noise. For a gemstone, every camera path that enters the stone must be traced through

[0]Source code available at: [GitHub repository].

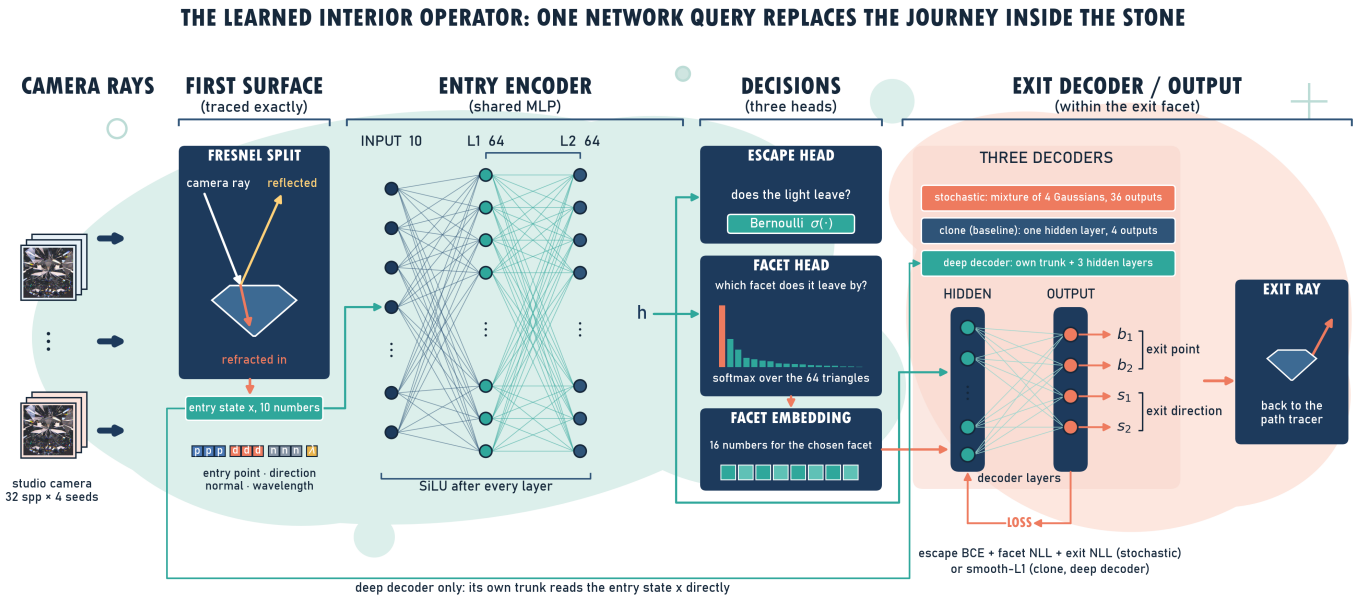


Fig. 1. The learned interior operator. Camera rays meet the stone's first surface, where Fresnel reflection or refraction is traced exactly. A refracted ray's entry state (position, direction, normal, wavelength; 10 numbers) passes through a shared two-layer encoder to an escape head, a facet head over the mesh's triangles, and a 16-number embedding of the chosen facet. One of three decoders then places the exit on that facet, and the exit ray returns to the path tracer. The deep decoder alone also reads the entry state through its own trunk.

all of its internal bounces, which is the work a learned operator aims to replace.

C. Neural appearance models and the RDM

A neural appearance model replaces part of the light transport with a small network evaluated at render time [4], [5]. In the cloth model [1], the multiple scattering inside a yarn is gathered into an RDM by firing rays at the fibre geometry and binning the light that leaves by outgoing direction for each incoming direction; small networks are fitted to the binned table and applied as a material. The present study starts from this design and adapts it to a diamond.

III. Related Work

A. Neural appearance and learned transport

Neural materials compress measured or simulated appearance into compact networks, from neural BRDFs with importance sampling [6] to multi-resolution neural materials [5] and real-time layered models [4]. These represent light that is reflected or scattered near a surface. Closer to this study, Kallweit et al. [7] replace multiple scattering in clouds with a radiance-predicting network, and Vicini et al. [8] learn where light leaves a translucent object given where it enters, using a conditional variational autoencoder and local shape descriptors. Both address diffusive media, whose exit distributions are smooth. Learned densities also serve as importance-sampling proposals [9]. This study targets the specular case these works exclude: a smooth dielectric whose exit is a discontinuous function of the entry state.

B. Rendering gemstones

Gemstone rendering has been approached with explicit optics: Sun et al. [10] model the refraction, reflection and dispersion of diamonds, and Guy and Soler [11] render gemstones in real time by tracing a bounded number of internal reflections. Spectral rendering with hero wavelengths [12] keeps dispersion affordable. These methods are exact or deliberately truncated; none learns the interior transport.

C. Representation limits of small networks

Multilayer perceptrons are biased towards smooth functions [13]. Fourier features [14] and multiresolution hash encodings [15] are standard remedies, and mixture density networks [16] represent several possible outcomes for one input, suiting an entry state that can escape by several internal routes. Moment-based filtering such as LEAN mapping [17] preserves variation that averaging would otherwise erase. Each of these is tested or adapted below.

D. Evaluating renders

Images are commonly compared with SSIM [18] or FLIP [19]. Both are sensitive to Monte Carlo noise, which matters when a learned method is compared with a noisy reference. This study instead measures agreement against the reference's own noise level, corrects correlations for noise in the manner of Spearman's disattenuation [20], and attaches block-bootstrap intervals [21], [22].

IV. Preliminary Study: The Radiance Map

The project began as a direct adaptation of the cloth model: an RDM gathered on a diamond mesh, compressed into small

networks and applied as a material on the stone's surface. Each step below removed one explanation for why this did not produce a diamond.

a) Is it a sampling problem?: Soh and Montazeri gather 50–100 million fully traced rays per yarn material until the map converges [1]. To give the RDM the same chance, it is gathered for each cut at five budgets from 2^{26} rays (67 million) to 2^{30} (1.07 billion) and rendered under the same rig (Fig. 2). If flat shading were sampling noise it would sharpen along this ladder; if it persists while the bins converge, the limit is what the map conditions on. *[Ladder renders pending.]*

b) Averaging erases the detail: Each bin stores the mean of the paths landing in it. In 99.8% of well-populated bins the standard deviation exceeds that mean (median ratio 2.53): more rays make the mean more precise but do not make the distribution narrower. A moment-based modification after LEAN mapping [17], which switches shading points on with probability $p = 1/(1 + r^2)$ at gain $1/p$ for relative spread r, restores the *amount* of sparkle but places it arbitrarily: the result reads as speckle.

c) The aim offset fixes coverage: The first gathers aimed every ray at the stone's centre, which caps the incidence angle each facet can receive: no checkpoint before the fix filled more than 41 of its 64 incident-direction bins. Firing each ray from a random point on a disc the size of the stone's silhouette (the *aim offset*), with directions stratified uniformly in $\cos\theta$ within each bin, fills all 64. Where a ray is aimed matters as much as its direction, a lesson carried into the operator's data (Sec. VII-F).

d) The missing variable is position: Asked to predict the exit triangle, a table conditioned on entry direction, as the RDM is, scores 0.068 with 32 cells and only 0.098 with 2,048, against 0.047 for the most common triangle; conditioning on the entry triangle instead reaches 0.192. Resolution is not the limit; the conditioning variable is. The learned interior operator follows: it keeps the RDM's idea of a learned interior but conditions on entry position, direction, normal and wavelength and returns a single exit rather than a binned average.

V. Methodology

A. Scene and reference

All renders use one studio scene: a perspective camera with a 30° field of view, a grey diffuse ground plane, textured area lights and two small intense sources. Structured lighting is essential, because under uniform lighting every exit direction returns the same radiance and errors become invisible. The *analytic reference* is a spectral path tracer in Mitsuba 3 [23] that treats the stone as a smooth dispersive dielectric (Sellmeier index), samples Fresnel reflection or refraction at every interaction and traces each interior path until escape or a depth of 128. Absorption is not modelled.

B. The learned interior operator

The entry state $\mathbf{x} \in \mathbb{R}^{10}$ holds position $\mathbf{p}$ (scaled by the mesh radius), interior direction $\boldsymbol{\omega}$, entry normal $\mathbf{n}$ and wavelength λ. With e the event that light escapes before the depth cap, f the exit triangle and $\mathbf{y} \in \mathbb{R}^4$ the exit coordinates, the operator models

$$p(e = 1, f, \mathbf{y} \mid \mathbf{x}) = s_\theta(\mathbf{x})\, \pi_\theta(f \mid \mathbf{x})\, q_\theta(\mathbf{y} \mid \mathbf{x}, f), \quad (1)$$

so non-escape is explicit rather than an artificial exit. Two coordinates are barycentric logits and two are tangent slopes on triangle f:

$$\boldsymbol{\beta} = \mathrm{softmax}(y_1, y_2, 0), \quad \boldsymbol{\omega}_{\mathrm{out}} = \frac{\mathbf{n}_f + y_3\mathbf{t}_f + y_4\mathbf{b}_f}{\|\mathbf{n}_f + y_3\mathbf{t}_f + y_4\mathbf{b}_f\|}, \quad (2)$$

with exit point $\sum_j \beta_j \mathbf{v}_{fj}$, so every prediction lies on the chosen triangle and points outward.

Fig. 1 shows the architecture. A two-layer, width-64 SiLU encoder [24] feeds an escape head and a facet head; a learned 16-number embedding of the chosen facet conditions the exit decoder. Three decoders share everything else: the *stochastic* decoder is a mixture of four Gaussians [16] sampled at render time (17,637 parameters on the round cut); the *clone* regresses a single exit with a smooth-L1 loss, after behavioural cloning [25] (15,557); and the *deep clone* uses three hidden layers and its own encoder (28,741). All three sample the exit facet; they differ only in how the exit is placed on it. Training minimises

$$\mathcal{L}_i = \mathrm{BCE}(s_\theta(\mathbf{x}_i), e_i) + e_i\,[-\log \pi_\theta(f_i \mid \mathbf{x}_i) + \ell_{\mathrm{coord}}(\mathbf{y}_i)] \quad (3)$$

with Adam [26].

C. Rendering with the operator

The first surface is traced exactly in both modes; only the journey inside the stone differs. When a camera path reaches the stone, reflection or refraction is sampled from the Fresnel coefficient. A reflected path continues outside as usual. A refracted path forms its entry state and joins a batch of up to 512 waiting entries, so the network is queried efficiently. For each entry the operator samples whether light escapes, the exit facet and the exit coordinates; the path then continues from the decoded exit as an ordinary scene ray. Non-escaping paths end, and exits that would immediately re-enter the stone are discarded. In analytic mode the same entry is instead traced through every internal bounce.

VI. Experiments

A. Experimental design

Five experiments test the hypotheses.

- **E1 Decoder comparison** (H1): stochastic, clone and deep-clone operators trained on the same paths for each of three cuts, rendered against the reference.
- **E2 Diagnostics** (H1): lookup baselines, a nearest-neighbour oracle over the training paths, an input ablation, and an *oracle-facet* render in which the exact tracer supplies the true exit facet.
- **E3 Recorded run** (H1, H2): one configuration (round cut, stochastic decoder, seed 23) re-run from gathered data to figures with every input recorded, trained to

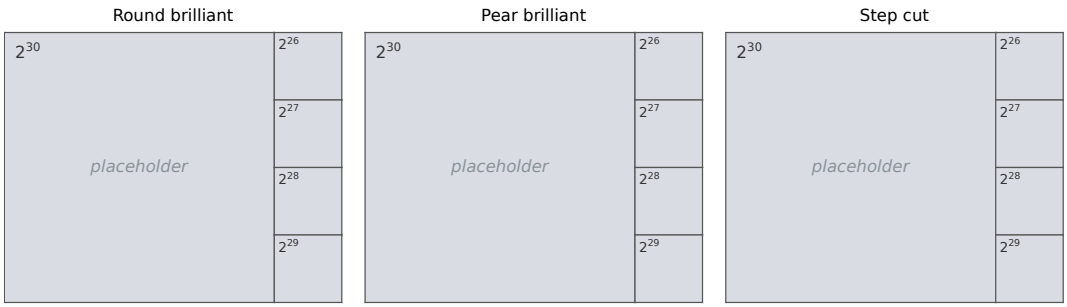


Fig. 2. RDM renders against gathering budget, after the 50–100 million rays per material of the cloth model [1]. For each cut the large panel uses 2^{30} rays and the stacked panels 2^{26} to 2^{29}; all use the same studio rig and render settings. *Placeholder layout: renders pending.*



Fig. 3. Round-cut renders from E1: stochastic, clone, deep clone and analytic reference at 128 spp. Deterministic placement restores sharp features but not their locations. Embedded scores use all seeds; Table II uses 64-spp halves.

TABLE I
EXPERIMENTAL PARAMETERS

Parameter	Value
Cuts (triangles / facets)	round 64/57, pear 62/33, step 112/50
Training pool per cut	1,048,576 paths from 65,536 entry states, split 80/20 by entry state
Held-out test pool (E3)	262,144 paths from 16,384 entry states, separate seed
Interior depth cap	128
Training	Adam, learning rate 10^{-3}, batch 256, clip 10
Stopping	E1: 80 (stochastic) / 30 (deep) epochs; E3: improvement $\leq 10^{-3}$ for 5 epochs
Renders	512×512, box filter, 4×32 spp per mode
Operator batch	512 entry states
Hardware (E3)	16-thread AMD CPU, 15 GB, Windows 11
Software (E3)	Mitsuba 3.9.1, PyTorch 2.10, Python 3.13

convergence and tested on an independent pool, broken down by path length and exit region.

- **E4 Cost** (H3): render and per-decision timing at equal time and equal quality, and the network's cost against batch size.
- **E5 Entry sampling**: camera-matched against uniform aim-offset training entries, on a 60-frame pear tumble.
- **E6 Transfer**: an operator trained on one cut, re-pointed at another without retraining: round (GIA) to pear, and pear to step cut.

B. Baselines

- **Analytic reference**: the exact path tracer at the same settings.
- **Lookup tables**: the most common exit triangle, and tables keyed on entry direction (the RDM's conditioning) or entry triangle.
- **Nearest-neighbour oracle**: the exits of the closest training entry states, a non-parametric test of whether the exit is predictable at all.

C. Evaluation metrics

Metrics use the pixels whose camera ray hits the stone, on linear luminance. Four seeds per mode are split into two independent halves; the reference compared with its own other half gives the *noise floor*, the agreement sampling noise alone allows.

- **Pixel correlation** $\rho(N, A)$ between neural and reference halves: is light in the right place?

TABLE II

IMAGE EVALUATION OF E1. FOUR 32-SPP SEEDS PER MODE; PIXEL CORRELATION, RELATIVE RMSE, HIGHLIGHT OVERLAP AND ENERGY USE 64-SPP HALVES, CONTRAST AND FACET CORRELATION THE FULL AVERAGE. "REFERENCE HALVES" IS THE NOISE FLOOR.

Cut	Decoder	Pixel corr. ↑	Rel. RMSE ↓	Top-1% overlap ↑	Energy error	Contrast	Facet corr. ↑
Round	Stochastic	0.152	3.750	0.197	−0.8%	0.93	0.399
	Clone	0.048	4.741	0.071	+26.4%	2.35	0.288
	Deep clone	0.038	4.849	0.117	+13.8%	2.74	0.038
	Reference halves	0.957	1.140	0.777	+0.6%	3.78	0.9996
Pear	Stochastic	0.136	1.834	0.007	+17.3%	0.89	0.645
	Deep clone	0.054	3.795	0.002	+47.8%	2.31	0.414
	Reference halves	0.899	0.684	0.595	+0.4%	1.48	0.9996
Step	Stochastic	0.293	2.353	0.059	+14.2%	0.84	0.880
	Deep clone	0.168	2.956	0.057	+15.3%	1.77	0.899
	Reference halves	0.942	0.817	0.650	−0.3%	2.37	0.9997

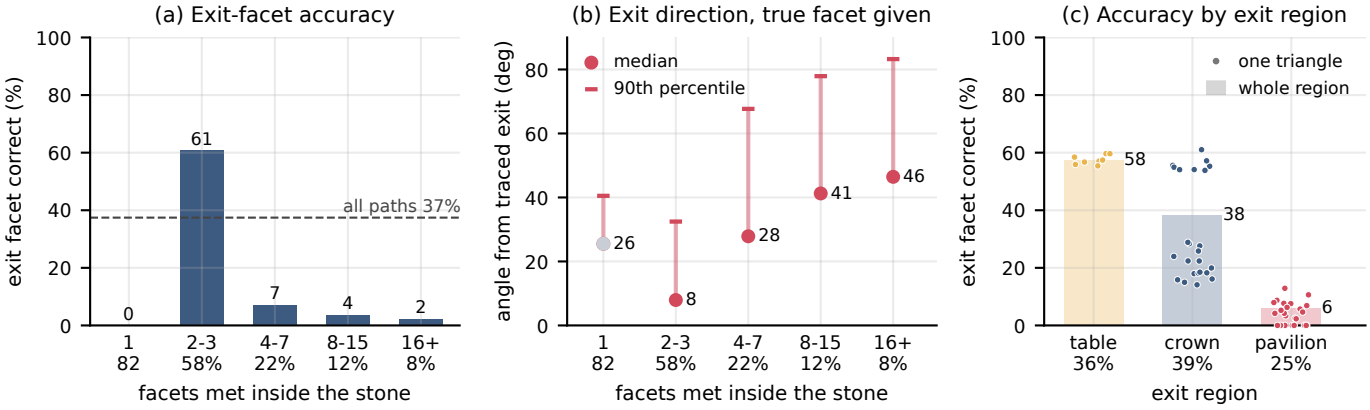


Fig. 4. Recorded run, 261,632 escaped held-out paths. (a) Exit-facet top-1 accuracy by the number of facets a path meets inside the stone; the second tick line is each group's share of paths. (b) Exit-direction error when the true facet is supplied: median and 90th percentile. (c) Share of exits through each region that the operator also sends there; bar: whole region, dot: one triangle (8 table, 24 crown, 24 of 32 pavilion triangles with at least 20 test exits).

TABLE III

ROUND-CUT EXIT-TRIANGLE PREDICTION ON HELD-OUT PATHS

Predictor	Top-1 accuracy
Most common triangle	0.047
Entry-direction table (32 / 2,048 cells)	0.068 / 0.098
Entry-triangle table	0.192
Nearest neighbour / majority of eight	0.287 / 0.343
Stochastic operator	0.380
Clone operator	0.410
Stochastic, recorded run, independent test pool	0.374

TABLE IV

RECORDED RUN: AGREEMENT WITH THE REFERENCE (ROUND, STOCHASTIC, 4×32 spp). BRACKETS: 95% INTERVAL; FLOOR: THE REFERENCE AGAINST ITS OWN OTHER HALF.

Measure	Whole stone	Interior only
Pixel correlation	0.141 [0.090, 0.235]	0.077 [0.063, 0.107]
noise floor	0.957	0.959
Disattenuated correlation	0.213	0.156
Rank / log correlation	0.32 / 0.33	0.21 / 0.23
Facet correlation	0.352 [0.03, 0.90]	0.246 [0.03, 0.76]
Energy error	−1.7%	−1.9%
Noise-free contrast (ref.)	0.77 (3.74)	0.59 (4.32)

- **Disattenuated correlation** $\rho/\sqrt{r_N r_A}$, dividing by each image's half-split reliability r: agreement with noise removed.
- **Facet correlation** across visible facet means: is coarse brightness right?
- **Energy error** $\epsilon_E = \sum N_p / \sum A_p - 1$: is the total right?
- **Contrast**, standard deviation over mean (noise-free version from the covariance of halves): is the image as sharp as the reference?
- **Interior-only** versions of each, using only light that passed through the operator (or, for the reference, the traced interior).
- **Operator metrics** on held-out paths: exit-facet top-1 accuracy, and the exit-direction angle when the true facet is supplied.

Ninety-five percent intervals come from bootstrapping 16×16-pixel blocks or facets.

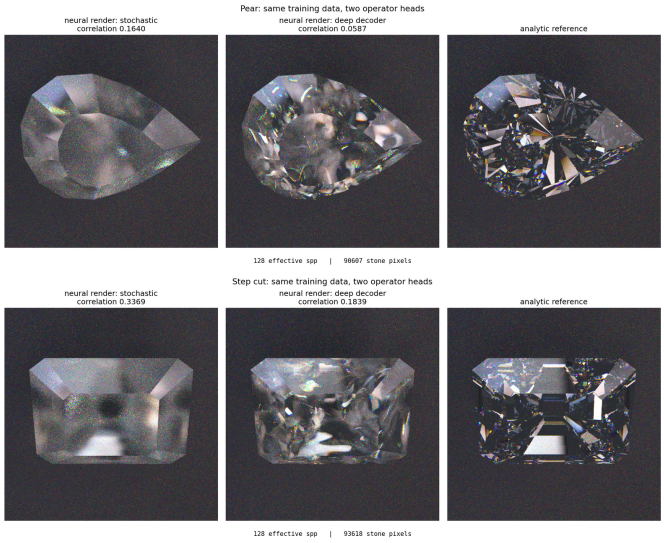


Fig. 5. Pear (top) and step cut (bottom) from E1: stochastic and deep-clone operators beside their references (128 spp). Broad shape and brightness survive on both cuts; the sharp internal reflections do not.

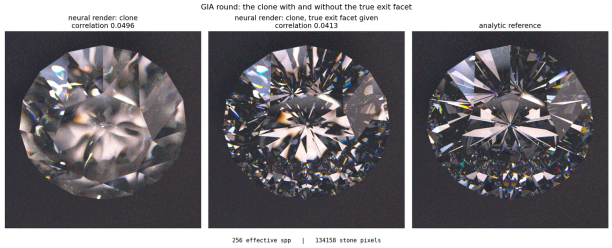


Fig. 6. Clone, clone supplied with the traced exit facet, and reference (256 spp). The correct facet restores contrast but not placement.

D. Data collection

Each recorded run writes, under one folder: the exact command and git commit, the software and hardware, hashes of the mesh, training data and checkpoint, per-epoch training and validation losses, per-seed render timing, every evaluation result as JSON and CSV, and the figures. The historical E1 runs predate this pipeline and record only their checkpoints and evaluations.

VII. Results and Analysis

A. The operator learns (E2)

a) Learning beats every lookup: The operators exceed every baseline (Table III); the clone is 43% above the nearest-neighbour oracle over a million training paths, so the network is not merely interpolating stored paths.

b) Position carries the information: With the architecture fixed, removing position from the input drops top-1 accuracy from 0.404 to 0.273, and an operator given only entry direction, the RDM's conditioning, falls to 0.117. Removing wavelength changes it by 0.0003.

TABLE V
Recorded cost, round cut, 512 × 512, 32 spp per render, mean of four seeds (renders run four at a time). Error is relative RMSE against the converged analytic image.

	Analytic	Neural
Render time (s)	**538**	737
Time per interior decision (μs)	63	**18**
Interior share of render time	49%	10%
Error after four renders	**0.56**	3.69
Error at equal time (2,950 s)	**0.48**	3.69
Lowest reachable error	$\to 0$	3.65
Time to reach the other's best	**51 s**	never
Training time	–	777 s
Model size	1 KB mesh	70 KB

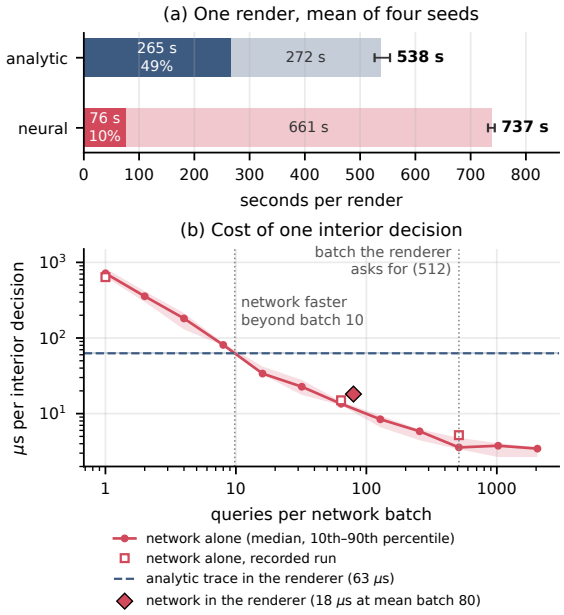


Fig. 7. Where the time goes (recorded run). (a) Render time split into interior decisions (dark) and everything else (light), with the seed range. (b) Cost of one interior decision against batch size: a standalone benchmark (curve, 10th–90th percentile band), the recorded run's benchmark (squares, different machine load), the analytic trace in the renderer (dashed) and the network in the renderer at its mean batch (diamond).

B. Sharper is not more faithful (E1, H1)

a) A more accurate operator agrees less: The deep clone predicts exit facets more accurately than the stochastic decoder on every cut (validation top-1 0.392 against 0.371 round, 0.338 against 0.311 pear, 0.328 against 0.305 step), and its images look more like a diamond, with contrast up to the reference's. Yet its pixel correlation is lower on every cut: 0.038 against 0.152, 0.054 against 0.136, and 0.168 against 0.293 (Figs. 3 and 5, Table II). The pattern holds on all three cuts, so it is not a property of one geometry. This supports H1.

b) The stochastic image is right on average, wrong in detail: The stochastic round image preserves total energy within 1% but has a quarter of the reference's contrast (0.93 against 3.78): its exits spread so widely that each pixel

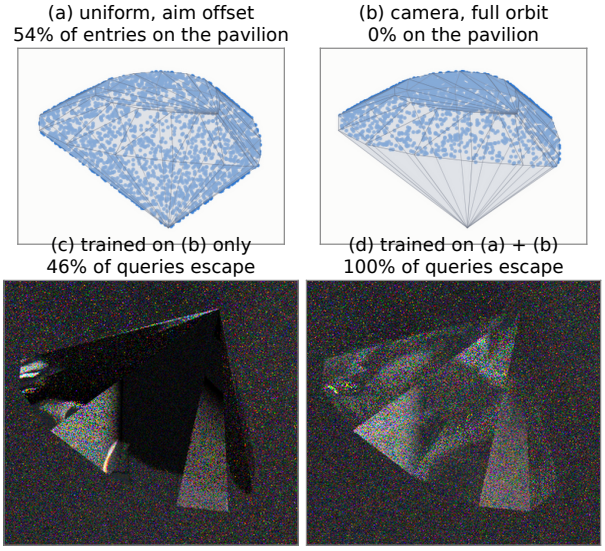


Fig. 8. Entry sampling on the pear. Top: where training entries land under (a) uniform directions with the aim offset and (b) camera rays from a full orbit. Bottom: a pavilion-facing tumble pose (frame 53 of 60) rendered by operators trained on (b) alone and on (a) and (b) together.

averages bright and dark parts of the lighting, like frosted glass.

c) Coarse structure can survive: Averaged over facets, the step cut's deep clone reaches 0.899, close to the reference's 0.9997, while its pixel correlation is 0.168: which facets are bright can be right while the sparkle within them is wrong.

C. Choosing the facet is not enough (E2)

a) The right facet restores the look, not the image: Supplying the clone with the true exit facet restores contrast (3.73 against 3.74) and the star pattern (Fig. 6), but correlation falls to 0.041. Within the correct facet the clone's exit is still about 24.7° from the truth, so sharp highlights land in the wrong places.

b) Blur and accuracy are separate axes: Decoding the stochastic model at its component means cuts the spread of its exits from 26.0° to 8.1°, yet correlation falls from 0.174 to 0.153. A blurred image keeps partial overlap with the reference; a sharp but misplaced one loses it.

D. Where the operator fails (E3, H2)

a) The earlier result reproduces: Trained to convergence (stopping at epoch 62, best 57) and tested on an independent pool, the operator reaches top-1 accuracy 0.374, matching its validation score of 0.375, and pixel correlation 0.141, matching the earlier 0.152 within its interval (Table IV). Under-training does not explain the result.

b) The disagreement is structure, not noise: Removing sampling noise (disattenuated 0.213) or the influence of a few bright pixels (rank 0.32, log 0.33) raises the score but leaves it far below the 0.957 floor.

c) Light through the network agrees least: The interior-only correlation, 0.077, is half the whole-stone value: untouched surface reflections lift the whole-stone score.

d) Accuracy collapses on long paths: Paths meeting two or three facets (58% of paths) are predicted correctly 61% of the time and leave within a median 8° of the traced direction (Fig. 4a,b). For paths meeting four or more facets (42%), accuracy falls to 2–7% and the median error grows to 28°–46°.

e) The pavilion fails despite ample data: Exits through the table are predicted correctly 58% of the time and through the crown 38%, but through the pavilion only 6%, although the pavilion carries a quarter of all exiting light (Fig. 4c). Its large triangles each have about 1.6×10^4 training exits yet score 2–13%, so the failure follows route length, not data. This supports H2.

E. Faster per decision, slower per image (E4, H3)

a) Each decision is faster: One learned interior decision costs 18 μs against 63 μs to trace it, 3.5 times faster, and the interior's share of the render drops from 49% to 10% (Table V, Fig. 7a).

b) The whole render is slower: A neural render takes 37% longer: the time outside the interior grows from 272 s to 661 s, spent gathering entry states, preparing inputs and decoding exits. Batch size limits the saving (Fig. 7b): one query alone costs about 0.7 ms, the network only overtakes tracing beyond about ten queries per batch, and although the renderer asks for 512, its batches average 80 as paths finish.

c) No budget reaches analytic quality: Error is measured against the converged analytic image. The neural error is a fixed bias plus noise and levels off at 3.65, whereas a single analytic render already reaches 1.12 and the error model, extrapolating below one render, places the analytic image at the neural floor after about 51 s. At equal time the analytic error is 7.7 times lower. This supports H3.

F. Entry sampling decides coverage (E5)

a) Camera sampling misses the pavilion: Camera-matched entries match what the renderer asks, but a camera above the stone sees only the crown, so no entries land on the pavilion; uniform sampling with the RDM's aim offset puts 54% there (Fig. 8). Trained on camera entries alone, the operator lets only 46% of queries escape when the pavilion faces the camera, and the pavilion renders nearly black.

b) Coverage repairs escape, not placement: Adding uniform aim-offset entries restores escape to 100% at every pose and lights the pavilion, but correlation at that pose stays at 0.018 against a floor of 0.951. The recorded run, trained on camera entries only, is therefore a front-view result.

G. Transfer across cuts (E6)

a) What transfer requires: The operator's facet head is a categorical over the triangles of the mesh it was trained on, so it cannot be applied to a new cut directly. Each target triangle is instead given the identity of the source triangle whose outward normal points most nearly the same way, and the head's weights and facet embeddings are permuted to match; nothing is retrained. This imposes three restrictions. First, the face counts differ: round to pear (64 to 62 triangles)

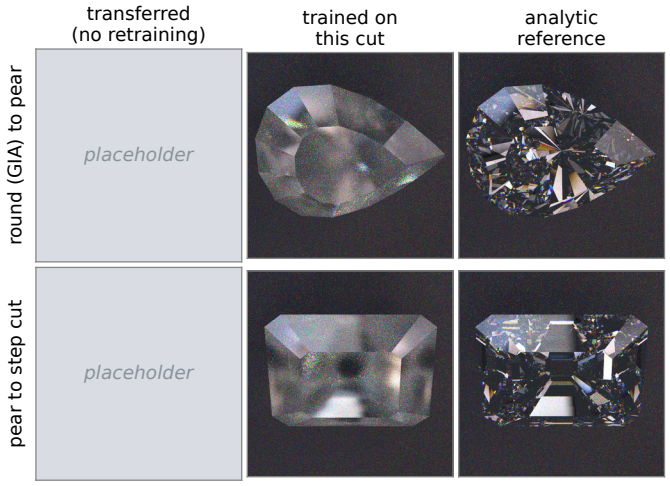


Fig. 9. Transfer across cuts (E6). Each row shows an operator trained on one cut and re-pointed at another without retraining, beside the stochastic operator trained on the target cut and the analytic reference (128 spp). *Transferred renders pending.*

drops two source classes, and pear to step cut (62 to 112) matches 62 triangles one to one and lets the other 50 share the probability of their best-aligned source triangle. Second, matching by normal ignores position, so facets that face the same way on different parts of the stone, such as a step cut's parallel steps, receive the same identity. Third, the encoder and decoder learned the internal routes of the source stone and are given no description of the new geometry, so transfer tests whether those learned routes carry over at all.

b) Earlier evidence: A pear operator re-pointed at a round variant (62 to 64 triangles) matched normals to within a median of $8.2°$, but a mean of $30.9°$ and a worst case of $138.9°$. Its whole-stone correlation of 0.37 came almost entirely from an untouched surface highlight: removing that patch dropped correlation to -0.001, and the operator component scored 0.004. Transfer is therefore judged on interior-only scores.

c) Round to pear, pear to step cut: Both transfers use the same render settings as the native operators of Fig. 5 (Fig. 9). *[Pending: interior-only and whole-stone correlation, energy error and mean normal alignment for each transfer, against the native operator's scores of 0.136 (pear) and 0.293 (step).]*

VIII. Discussion

A. Why accuracy does not reach the image

For entry state $\mathbf{x}$ and exit state z, the light carried back is

$$L(\mathbf{x}) = \int L_{\text{ext}}(z)\, p(z \mid \mathbf{x})\, \mathrm{d}z. \tag{4}$$

The training loss weights every exit equally, but the image weights each exit by the radiance L_{ext} it reaches: a small angular error at the edge of a bright lamp matters greatly, a large one between dark directions hardly at all. Facet accuracy and mean angular error are therefore incomplete proxies for the image. A broad exit distribution spreads energy over correct and incorrect regions and keeps some overlap; a sharp but misplaced one loses it, which is why every step towards sharpness in E1 and E2 moved correlation down. This explanation is consistent with the results but untested; a radiance-weighted objective would test it.

The breakdown in E3 adds that accuracy follows route length, not data. A long route changes facet whenever the path crosses any internal facet edge, so its exit varies rapidly with the entry state, and a small smooth network blends across such changes [13]. Local encodings [15] are a natural next test.

B. Limitations

The study uses one lighting rig, mostly one view, simplified meshes and a facet head tied to its mesh. The full recorded pipeline has run for one configuration; the E1 results use single training seeds and fixed epoch budgets, and multi-seed runs for all cuts and decoders, the RDM ladder and the transfer renders are pending. Renders ran four at a time, so absolute times include contention, and ground caustics are not evaluated.

IX. Conclusion

This study adapted a neural appearance model from cloth to faceted gemstones and generalised its direction-conditioned radiance map into a position-aware learned interior operator.

The operator learns: it predicts exit facets better than every lookup baseline, including a nearest-neighbour oracle over a million paths, and generalises to an independent test pool (top-1 0.374).

Accuracy does not transfer to the image (H1). On every cut the more accurate deep decoder renders a sharper diamond that agrees with the reference less (pixel correlation 0.038–0.168 against 0.136–0.293), and even the true exit facet leaves correlation at 0.041.

The failure is localised (H2). Exit facets are right for 61% of short paths but at most 7% of paths meeting four or more facets, and for 6% of pavilion exits despite ample training data.

The operator does not save time (H3). Each decision is 3.5 times faster, yet renders are 37% slower and the image settles at an error a single analytic render beats.

Future work should test radiance-weighted training and local encodings on the long-route cases, and repeat the recorded runs across seeds, cuts and lighting.

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